Ravi Raj

Greater Noida, India | +91-91421-20466 | raviroy2002@gmail.com | linkedin.com/in/raviraj12 | github.com/ravicoder01

Summary

AI/ML developer (B.Tech CSE – AI & ML, CGPA 8.55/10) who builds agentic RAG systems and fine-tunes open-source LLMs. Fine-tuned 3 LLMs with QLoRA during a 3-month AI internship, reaching 97.14% agent-routing accuracy, and built and deployed an agentic RAG assistant over Indian law. Seeking AI/ML, GenAI and NLP internships.

Experience

AI Developer Intern Jul 2026 – Oct 2026

L-VelUp Edulab FZCO, Dubai | GitHub *Remote*

- Created a **5,000-record dataset** for fine-tuning an in-house routing model for Stanza AI, a multi-agent LLM assistant for hotel operations, aimed at replacing third-party LLM APIs.
- Fine-tuned **3 open-source LLMs** (Qwen2.5-7B, Mistral-7B-v0.3, Phi-4-mini 3.8B) with **QLoRA** (4-bit, r=16, α=32) on a single T4 GPU, using a balanced 700-record set across **7 specialist agents** (560/70/70 split).
- Qwen2.5-7B achieved **97.14% agent-routing** and **88.57% exact agent + intent accuracy** on the held-out test set, beating Mistral-7B (84.29%) and Phi-4-mini (78.57%); recommended it as the base model for the next phase.
- Researched dataset strategy, evaluation metrics and single-model vs per-agent architectures; presented findings in regular engineering syncs.

Projects

LegalEase – Agentic RAG Legal Assistant | *LangGraph, FAISS, LLaMA-3.3* | GitHub | Live Jun 2026

- Indexed **3 Indian legal acts** (POCSO, RTI, Consumer Protection Act) into **309 FAISS vectors** using Hugging Face embeddings.
- Engineered a LangGraph agent with tool-based retrieval, persistent conversational memory and automatic routing across the 3 acts, removing manual topic selection.
- Generated **3 structured outputs** with LLaMA 3.3-70B: plain-language summaries, JSON key-information extraction and audio narration.
- Resolved production bugs in text encoding, LLM output parsing and retrieval metadata filtering; deployed on Streamlit Cloud.

Credit Card Fraud Detection | *XGBoost, SMOTE, Scikit-learn* | GitHub Feb 2026

- Trained an XGBoost classifier on **284,807 transactions** and handled severe class imbalance with SMOTE, achieving **91% recall** on fraud cases.

Sentiment Analysis – LSTM | *TensorFlow, Keras, Streamlit* | GitHub | Live Mar 2026

- Built an end-to-end NLP pipeline (tokenisation, padding, embedding + LSTM) on **50,000 IMDB reviews**, reaching **76% test accuracy**; deployed on Streamlit Cloud.

Technical Skills

Languages: Python, SQL, Java, C
GenAI & LLMs: LangChain, LangGraph, Agentic RAG, FAISS, Hugging Face Transformers, QLoRA/LoRA, PEFT, Prompt Engineering, LangSmith, Groq API
ML & DL: Scikit-learn, XGBoost, TensorFlow, Keras, RNN/LSTM, NLP, Feature Engineering, SMOTE, Hyperparameter Tuning
Tools & Deployment: Streamlit, Flask, Streamlit Cloud, Render, Git, GitHub, Google Colab, Jupyter

Education

IILM University Greater Noida, UP

B.Tech, Computer Science & Engineering (AI & ML) – CGPA: 8.55/10 *2023 – 2027*

Certifications & Achievements

- **Bronze Medal**, IIC 2025 · **GSSoC 2025** Contributor · LeetCode 100-day streak
- Machine Learning Specialization (DeepLearning.AI) · Apply AI & Modern AI (Cisco) · Responsible AI, LLMs & GenAI (Google Cloud)